

SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI - 600 063
PERIODIC TEST - 1 (2026 - 2027)
CHEMISTRY
SET -3

CLASS : XII
DATE : 11.06.2026

MAX. MARKS : 20
DURATION : 40 mins.

Q.NO	CHOOSE THE MOST APPROPRIATE ANSWER FROM THE FOLLOWING OPTIONS:										
1	If  react with Cl_2 in presence of light and then react with Na metal in dry ether, final product is: a)  b)  c)  d) 										
2	Chloroform is stored in closed dark coloured bottles completely filled with it because: a) It is slowly reduced by air in the presence of light to carbonyl chloride. b) It is slowly oxidised by air in the absence of light to carbonyl chloride. c) It is slowly reduced by air in the absence of light to carbonyl chloride. d) It is slowly oxidised by air in the presence of light to carbonyl chloride.										
3	Match the alkyl halide reacting with reagent given in column I with their respective substituted product given in column II and choose the correct option from the codes given below. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Column I (Reagent)</th><th>Column II (Substituted product)</th></tr> </thead> <tbody> <tr> <td>A. KCN</td><td>1. RH</td></tr> <tr> <td>B. AgCN</td><td>2. RCN</td></tr> <tr> <td>C. KNO_2</td><td>3. RNC</td></tr> <tr> <td>D. LiAlH_4</td><td>4. R-O-N=O</td></tr> </tbody> </table> a) A-2, B-3, C-1, D-4 b) A-3, B-2, C-1, D-4 c) A-2, B-3, C-4, D-1 d) A-3, B-2, C-4, D-1	Column I (Reagent)	Column II (Substituted product)	A. KCN	1. RH	B. AgCN	2. RCN	C. KNO_2	3. RNC	D. LiAlH_4	4. R-O-N=O
Column I (Reagent)	Column II (Substituted product)										
A. KCN	1. RH										
B. AgCN	2. RCN										
C. KNO_2	3. RNC										
D. LiAlH_4	4. R-O-N=O										
4	If the boiling point of 2-methyl-2-bromopropane is 346K, predict the boiling point of n-Butyl bromide? a) 300K b) 325K c) 350K d) 375K										
5	The number of optically active stereoisomers possible for butane - 2,3 - diol are: a) 1 b) 2 c) 3 d) 0										
6	Which of the following is not an allylic halide? a) 5-Bromopent-1-ene b) 4-Bromopent-2-ene c) 3-Bromo-2-methylbut-1-ene d) 1-Bromobut-2-ene										
7	The IUPAC name of tertiary butyl bromide is: a) 3-Bromobutane b) 2-Bromo-2-methylpropane c) 1,2-bromobutane d) 4-bromobutane										
8	Arrange the following compounds in the increasing order of their dipole moment values: I. Toluene II. m-Dichlorobenzene III. o-Dichlorobenzene IV. p-Dichlorobenzene a) I<IV<II<III b) IV<I<II<III c) IV<I<III<II d) IV<II<I<III										
9	Select the incorrect statement regarding $\text{S}_{\text{N}}2$ reaction mechanism. a) As the reaction progresses and the bond between the incoming nucleophile and the carbon atom starts forming, the bond between carbon atom and leaving group weakens. b) There is the inversion of configuration. c) The order of reactivity of alkyl halides is $3^\circ < 2^\circ < 1^\circ$. d) Carbon atom is bonded to four atoms during the transition state.										

10	In the iodination of benzene, which compound causes the reduction of iodobenzene, back to benzene? a) Iodic acid ☒ b) Hydrogen iodide c) Nitric acid d) Mercuric oxide
	For the given Assertion(A) and Reason(R) in the questions 11 -15, Select the most appropriate answer from the options given below: a) Both A and R are true, and R is the correct explanation of A. b) Both A and R are true, and R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true. e) Both A and R are false.
11	Assertion(A): tertiary butyl bromide undergoes Wurtz reaction to give 2,2,3-trimethylbutane. Reason(R): In Wurtz reaction, alkyl halides react with sodium in dry ether to give hydrocarbon containing double the carbon atoms that are present in the halide.
12	Assertion(A): Presence of nitro group decrease the nucleophilic substitution reaction in aryl halides. Reason(R): The intermediate carbocation is stabilized due to the presence of nitro group.
13	Assertion(A): Haloalkanes are insoluble in water. Reason(R): Intermolecular attractions between haloalkanes and water molecules have much more strength than intermolecular attractions between water molecules.
14	Assertion(A): The boiling points of alkyl halides decrease in the order: RI > RBr > RCl > RF Reason(R): The boiling points of alkyl halides are considerably higher than those of the hydrocarbon of comparable molecular masses.
15	Assertion(A): Yield of any single alkyl halide is low in free radical halogenation of alkanes. Reason(R): Free radical halogenation of alkanes leads to the formation of mixture of isomeric mono and polyhaloalkanes.
16	What is the major product formed when chlorobenzene reacts with Concentrated nitric acid and Concentrated sulphuric acid? a) Nitrobenzene b) 1-Chloro-2-nitrobenzene c) 1-Chloro-3-nitrobenzene ☒ d) 1-Chloro-4-nitrobenzene
17	Predict the major product when $C_6H_5-CH=CH-CH_3$ is treated with HBr. ☒ a) 1-bromo-1-phenylpropane ☒ b) 2-bromo-1-phenylpropane c) 2-bromo-2-phenylpropane d) 1-bromo-2-phenylpropane
18	What is the reason for iodoform to be used as an antiseptic? a) Due to its unpleasant odour b) Due to its melting point c) Due to its solubility in alcohol ☒ d) Due to the liberation of free iodine
19	Which of the following is the suitable medium for preparing Grignard reagents? a) Dry acetone ☒ b) Dry ether c) Concentrated HCl d) Alcoholic KOH
20	Predict the minor product formed when chlorobenzene reacts with chloromethane in the presence of anhydrous $AlCl_3$? a) Toluene b) m-Chlorotoluene ☒ c) o-Chlorotoluene d) p-Chlorotoluene

CLASS: XII
DATE: 24.07.2026

MAX. MARKS: 35
DURATION: 1hr. 30 min.

GENERAL INSTRUCTIONS:

Read the following instructions carefully and follow them:

- (i) This question paper contains 16 questions. All questions are compulsory.
- (ii) This question paper is divided into five sections – Section A, B, C, D and E.
- (iii) Section A – questions number 1 to 8 are multiple choice type questions. Each question carries 1 mark.
- (iv) Section B – questions number 9 and 10 are very short answer type questions. Each question carries 2 marks.
- (v) Section C – questions number 11 to 13 are short answer type questions. Each question carries 3 marks.
- (vi) Section D – question number 14 is a case-based question carrying 4(2+1+1) marks.
- (vii) Section E – questions number 15 and 16 are long answer type questions. Each question carries 5 marks.
- (viii) There is no overall choice given in the question paper.

SECTION A

1. The colligative property preferred for the determination of molar mass of polymers is:
(A) Depression in freezing point (B) Osmotic pressure
(C) Relative lowering in vapour pressure (D) Elevation in boiling point
2. A conductivity cell contains electrodes made up of
(A) Silver (B) Copper (C) Platinum (D) Zinc
3. A compound $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ undergoes complete dissociation in water. The Van't Hoff factor is
(A) 9 (B) 6 (C) 4 (D) 3
4. Which of the following cells has a constant voltage throughout its life?
(A) Dry cell (B) Daniell cell (C) Electrolytic cell (D) Mercury cell
5. The standard electrode potential of 'A', 'B' and 'C' are + 0.68 V, - 2.54 V and - 0.50 V respectively. The order of their reducing power is
(A) $A > B > C$ (B) $A > C > B$ (C) $B > C > A$ (D) $C > B > A$
6. ΔG° and E°_{cell} for a spontaneous reaction will respectively be
(A) positive and positive (B) positive and negative
(C) negative and positive (D) negative and negative

For Questions number 7 and 8, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 - (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
 - (C) Assertion (A) is true, but Reason (R) is false.
 - (D) Assertion (A) is false, but Reason (R) is true.
7. **Assertion (A):** When glucose is added to water, an elevation in boiling point is observed.
Reason (R): The lowering of vapour pressure causes elevation in the boiling point.
8. **Assertion (A):** For strong electrolytes, there is a slow increase in molar conductivity with dilution and can be represented by $\Lambda_m^\circ = \Lambda_m - A c^{1/2}$ the equation
Reason (R): The value of the constant 'A' for NaCl, CaCl_2 , and MgSO_4 in a given solvent and at a given temperature is different.

SECTION B

9. Define azeotropes. What type of azeotrope is formed by negative deviation from Raoult's law?
10. State Kohlrausch's Law of independent migration of ions. Solutions of two electrolytes 'A' and 'B' are diluted. The limiting molar conductivity of 'B' increases 2.5 times while that of 'A' increases 25 times. Which of the two is a strong electrolyte? Justify your answer.

SECTION C

11. The conductivity of the 0.001 M solution of CH_3COOH is $3.905 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its molar conductivity and degree of dissociation.
(Given: $\lambda_{\text{m}}^0(\text{CH}_3\text{COO}^-) = 40.9 \text{ S cm}^2 \text{ mol}^{-1}$, $\lambda_{\text{H}^+}^0 = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$)
12. Calculate emf of the following cell at 298 K:
 $\text{Mg(s)} | \text{Mg}^{2+} (0.001 \text{ M}) || \text{Cu}^{2+} (0.0001 \text{ M}) | \text{Cu(s)}$ (Given: $E^\circ_{\text{cell}} = 2.71 \text{ V}$, $\log 10 = 1$)
13. The vapour pressure of two pure liquids A and B at 400 K are 450 and 700 mmHg respectively. Find out the composition of liquid mixture if the total vapour pressure of mixture is 600 mmHg. Also find the composition in vapour phase.

SECTION D

14. Read the following passage and answer the questions that follow:

Henna is investigating the melting point of different salt solutions. She makes a salt solution using 10 mL of water with a known mass of NaCl salt. She puts the salt solution into a freezer and leaves it to freeze. She takes the frozen salt solution out of the freezer and measures the temperature when the frozen salt solution melts. She repeats each experiment and tabulates the data as follows:

S.No	Mass of the salt used in g	Melting point in °C	
		Readings Set 1	Reading Set 2
1	0.3	-1.9	-1.9
2	0.4	-2.5	-2.6
3	0.5	-3.0	-5.5
4	0.6	-3.8	-3.8
5	0.8	-5.1	-5.0
6	1.0	-6.4	-6.3

Assuming the melting point of pure water as 0 °C, answer the following questions:

- One temperature in the second set of results does not fit the pattern. Which temperature is that? Justify your answer.
- Why did Henna collect two sets of results?
- In place of NaCl, if Henna had used glucose, what would have been the melting point of the solution with 0.6 g glucose in it?

SECTION E

15. (i) Boiling point of water at 750 mmHg is 99.48°C. How much sucrose ($M = 342 \text{ g mol}^{-1}$) is to be added to 500 g of water such that it boils at 100°C ?
[K_b for water = $0.52 \text{ K kg mol}^{-1}$]
- (ii) Define the following terms: (a) Ideal Solution (b) Osmotic Pressure
16. (i) Write the electrodic reactions that take place in a fuel cell.
- (ii) Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10.
- (iii) Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100 Ω . If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520 Ω , calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29 S/m.

SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI - 600 063
I TERM EXAMINATION (2026 - 2027)
CHEMISTRY

CLASS: XII

DATE : 17.09.2026

MAX. MARKS: 70

DURATION : 3 Hrs.

General Instructions:

- a) There are 33 questions in this question paper with internal choice.
- b) Section A consists of 16 objective type questions carrying 1 mark each.
 - Q. No.1 to 12 are Multiple choice questions.
 - Q. No.13 to 16 are Assertion-Reason type questions.
- c) Section B: Q. No. 17 to 21 are short answer questions and carry 2 marks each.
- d) Section C: Q. No. 22 to 28 are short answer questions and carry 3 marks each.
- e) Section D: Q. No. 29 and 30 are case - based questions carrying 4 marks each.
- f) Section E: Q. No. 31 to 33 are long answer questions carrying 5 marks each.
- g) All questions are compulsory.

SECTION - A

The following questions (1-12) are multiple-choice questions with one correct answer. Each question carries 1 mark. There is no internal choice in this section.

1. Which type of isomerism is shown by the complexes $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$?
 (a) Linkage (b) Ionisation (c) Optical (d) Solvate
2. If molality of a dilute solution is doubled, the value of the molal elevation constant (K_b) will be
 (a) halved (b) doubled (c) tripled (d) unchanged
3. HCN is a toxic gas used in qualitative analysis. If the solubility of HCN in water at STP is 0.175 ^{molal}, what is the value of K_H ? [Partial pressure of HCN at STP = 0.987 bar]
 (a) 318 bar (b) 300 bar (c) 101 bar (d) 423 bar
4. Oxidation state of central metal atom in the given complex is : $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_3$
 (a) +2 (b) +3 (c) +1 (d) +4
5. The core atom of which of the following biologically significant coordination molecules is magnesium?
 (a) Vitamin B₁₂ (b) Haemoglobin (c) Chlorophyll (d) Carboxypeptidase-A
6. Choose the compound which is more acidic than phenol :
 (a) o-nitrophenol (b) ethanol (c) o-methylphenol (d) p-methoxyphenol
7. Which of the following aqueous solutions has the highest vapour pressure at 300 K?
 (a) 1MNa₃PO₄ (b) 1MCaCl₂ (c) 1MKNO₃ (d) 1MC₆H₁₂O₆
8. The correct IUPAC name of $(\text{CH}_3)_3\text{C}-\text{CH}_2\text{Br}$ is :
 (a) 2,2-Dimethyl-2-bromopropane (b) 1-Bromo-2,2,2-trimethylethane
 (c) 2-Bromo-1,1,1-trimethylethane (d) 1-Bromo-2,2-dimethylpropane
9. Which of the following is most reactive towards nucleophilic substitution reaction?
 (a)  (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ (c) $\text{CH}_2=\text{CHCl}$ (d) $\text{CH}_2=\text{CHCH}_2\text{Cl}$
10. The suitable Grignard reagent used for the synthesis of $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$ with methanal is :
 (a) $(\text{CH}_3)_2\text{CHCH}_2\text{MgBr}/\text{H}_2\text{O}$ (b) $\text{CH}_3\text{MgBr}/\text{H}_2\text{O}$
 (c) $(\text{CH}_3)_2\text{CHMgBr}/\text{H}_2\text{O}$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}/\text{H}_2\text{O}$

11. The correct order for the increasing oxidizing power in 3d series of transition elements is :

- (a) $\text{MnO}_4^- < \text{Cr}_2\text{O}_7^{2-} < \text{VO}_2^+$
(b) $\text{Cr}_2\text{O}_7^{2-} < \text{VO}_2^+ < \text{MnO}_4^-$
(c) $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$
(d) $\text{VO}_2^+ < \text{MnO}_4^- < \text{Cr}_2\text{O}_7^{2-}$

12. $(\text{CH}_3)_2\text{CHOCH}_3$ reacts with HI to give :

- (a) $(\text{CH}_3)_2\text{CHOH} + \text{CH}_3\text{OH}$ (b) $(\text{CH}_3)_2\text{CHOH} + \text{CH}_3\text{I}$
(c) $(\text{CH}_3)_2\text{CHI} + \text{CH}_3\text{OH}$ (d) $(\text{CH}_3)_2\text{CHI} + \text{CH}_3\text{I}$

The following questions (13-16) are Assertion - Reason type questions. Choose the correct answer out of the following choices.

- a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
c) Assertion is correct statement but reason is wrong statement.
d) Assertion is wrong statement but reason is correct statement.

13. **Assertion:** Transition metals like Fe, Cr and Mn form oxoions

Reason: Oxygen is highly electronegative and has a tendency to form multiple bonds.

14. **Assertion :** Phenol is more reactive than benzene towards electrophilic substitution reaction.

Reason : In the case of phenol, the intermediate carbocation is more resonance stabilized.

15. **Assertion:** Like bromination of benzene, ^{phenol} is also carried out in the presence of Lewis acid.

Reason: Lewis acid polarizes the bromine molecule.

16. **Assertion:** Ethers behave as bases in the presence of mineral acids.

Reason: Due to the presence of lone pairs of electrons on oxygen.

SECTION B

The following questions (17 - 21) are short answer type and carry 2 marks each.

17. Differentiate the following:

- a) Homoleptic and Heteroleptic complexes. b) Double salt and Complex salt.

18A. Account for the following:

- a) The melting and boiling points of Zn, Cd and Hg are low.
b) Transition metals and their many compounds acts as good catalysts.

OR

18B. a) Transition metals and their many compounds show paramagnetic behaviour.

b) Predict which of the following will be coloured in aqueous solution and why?

$\text{Sc}^{3+}, \text{Fe}^{3+}, \text{Zn}^{2+}$ [Atomic number : Sc = 21, Fe = 26, Zn = 30]

19. a) What is meant by 'disproportionation' of an oxidation state? Give an example.

b) Why the lowest oxide of transition metal is basic, the highest is amphoteric/acidic?

20. $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic though both are tetrahedral. Why?

21. State Raoult's law for a solution containing volatile components. Name the solution which follows Raoult's law at all concentrations and temperatures.

SECTION C

The following questions (22 - 28) are short answer type and carry 3 marks each.

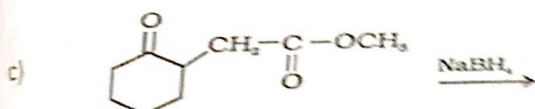
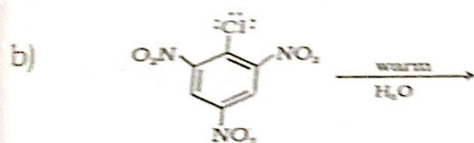
22. a) Which of the 3d series of transition metals exhibits the largest number of oxidation states and why?
 b) Actinoid contraction is greater from element to element than lanthanoid contraction. Why?
 c) In what way is the oxidation state of the transition elements different from that of the non-transition elements?
23. a) Explain the violet colour of the complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ on the basis of crystal field theory.
 b) What is meant by the chelate effect? Give an example.
 c) How many ions are produced from the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ in solution?
24. Calculate the freezing point of solution when 1.9 g of MgCl_2 ($M=95 \text{ g mol}^{-1}$) was dissolved in 50g of water, assuming MgCl_2 undergoes complete ionization. (K_f for water = $1.86 \text{ K kg mol}^{-1}$)
25. Calculate the depression in the freezing point of water when 10 g of $\text{CH}_3\text{CH}_2\text{CHClCOOH}$ is added to 250g of water. ($K_a = 1.4 \times 10^{-3}$, $K_f = 1.86 \text{ K kg mol}^{-1}$)
- 26A. Explain the following.
 (i) Kolbe's reaction. (ii) Reimer-Tiemann reaction. (iii) Williamson ether synthesis.

OR

- 26B. Explain the following.
 (i) Finkelstein reaction. (ii) Sandmeyer's reaction. (iii) Dow's process.
27. Explain the mechanism of the following reaction:



28. Complete the following:



SECTION- D

29. The following questions are case -based questions. Each question has an internal choice and carries 4 (1+1+2) marks each. Read the passage carefully and answer the questions that follow.

The transition metals form a large number of complex compounds in which the metal atoms are bound to a number of anions or neutral molecules by sharing of electrons. In modern terminology such compounds are called coordination compounds. The chemistry of coordination compounds is an important and challenging area of modern inorganic chemistry. New concepts of chemical bonding and molecular structure have provided insights into the functioning of these compounds as vital components of biological systems. Chlorophyll, haemoglobin and vitamin B_{12} are coordination compounds of magnesium, iron and cobalt respectively. Variety of metallurgical processes, industrial catalysts and analytical reagents involve the use of coordination compounds. Coordination compounds also find many applications in electroplating, textile dyeing and medicinal chemistry.

Answer the following questions:

- a) Identify the type of isomerism exhibited by the following complexes:
(i) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]^{2+}$ (ii) $[\text{Co}(\text{en})_3]\text{Cl}_3$
- b) (i) Write the formula of Iron(III) hexacyanidoferrate(II)
(ii) Write the IUPAC name of $[\text{Co}(\text{H}_2\text{O})(\text{CN})(\text{en})_2]^{2+}$

OR

- c) (i) Write the formula of Pentaamminenitrito-N-cobalt(III)
(ii) Write the IUPAC name of $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{NO}_2)]\text{Cl}$

30. Read the given passage and answer the questions that follow:

Alcohols play a very important role in our daily life. Ordinary spirit used as an antiseptic contains methanol. Ethanol is present in cough syrups, tonics, wine, beer, and whisky. Sugar, starch, cellulose are carbohydrates that also contain a large number —OH groups. Phenol is also an antiseptic in low concentration (0.2%) whereas a 2% solution of phenol is used as a disinfectant. The fragrance of rose is due to citronellol (unsaturated alcohol). Phenol is used for the preparation of many useful compounds like aspirin, methyl salicylate (Iodex), and phenyl salicylate (salol) used as an intestinal antiseptic.

Answer the following questions:

- a) How is phenol prepared from cumene?
b) How is phenol converted into benzoic acid?
c) Distinguish between phenol and benzyl alcohol?
d) Salicylic acid heated with acetic anhydride to form a compound which is very useful drug which lower the body temperature as well as use to prevent heart attack. What is the chemical name of compound?

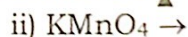
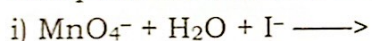
OR

- e) In Kolbe's reaction instead of phenol, phenoxide ion is treated with CO_2 . Why?

SECTION - E

The following questions are long answer type and carry 5 marks each. All questions have an internal choice.

31A. a) Complete the reaction mentioning all the products formed:



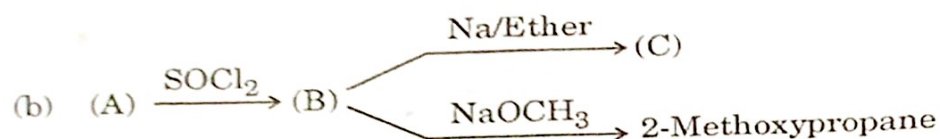
- b) When MnO_2 is fused with KOH in the presence of KNO_3 as an oxidizing agent, it gives a dark green compound (A). Compound (A) disproportionates in acidic solution to give purple compound (B). An alkaline solution of compound (B) oxidises KI to compound (C) whereas an acidified solution of compound (B) oxidises KI to (D). Identify (A), (B), (C), and (D).
- c) What is misch metal? Give one application.

OR

31B. a) Describe the oxidising action of potassium dichromate and write the ionic equations for its reaction with: (i) iodide (ii) iron(II) solution

- b) When chromite ore FeCr_2O_4 is fused with NaOH in presence of air, a yellow coloured compound (A) is obtained which on acidification with dilute sulphuric acid gives a compound (B). Compound (B) on reaction with KCl forms an orange coloured crystalline compound (C). Identify the compounds (A), (B) and (C) and explain with the equations.

32A. (i) Complete the following:



(ii) Account for the following:

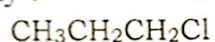
- Benzyl chloride is highly reactive towards $\text{S}_{\text{N}}1$ reaction.
- Butan-1-ol optically inactive but Butan-2-ol is optically active.
- Chloroform is stored in closed dark coloured bottles.

OR

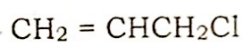
32B.(i) How do you convert the following :

- Propene to Propan-2-ol
- Ethanol to Ethanal
- Anisole to 2-Methoxyacetophenone

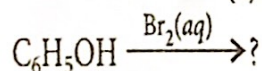
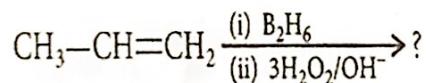
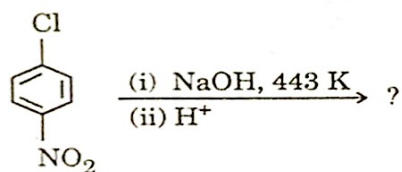
(ii) Which one of the following pairs of substances undergoes $\text{S}_{\text{N}}1$ reaction faster and why ?



OR



33A. a) Write the major product in the following:



b) Give reasons for the following:

- Chlorine is ortho/para directing in electrophilic aromatic substitution reactions, though chlorine is an electron withdrawing group.
- Racemic mixture is optically inactive.

OR

33B. a) How would you obtain the following:

- Benzene to 4-bromonitrobenzene
- 1-bromopropane to 2-bromopropane

b) Write the structure of the products when Butan-2-ol reacts with the following:

- CrO_3
- SOCl_2

c) Differentiate between $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ mechanisms and give examples.
